

YUG UPADHYAY

Full-Stack Engineer · Cloud-Native & AWS · AI-Augmented Development

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SUMMARY

Production-focused Full-Stack Engineer building and shipping scalable, cloud-native SaaS applications using MERN, Next.js, and AWS — from zero to production. Combines strong backend architecture with AI integration (LLMs, OpenAI API, prompt engineering) and hands-on AWS cloud infrastructure to deliver secure, high-performance systems. Accelerates development cycles through AI-augmented workflows and a security-first, scalable-architecture mindset.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Java, C/C++, SQL, R

Frontend: React.js, Next.js, Redux, TailwindCSS, ShadCN UI, HTML5, CSS3

Backend: Node.js, Express.js, REST API Design, JWT Authentication, WebSockets

Databases: MySQL, MongoDB, Firebase (Firestore)

Cloud & AI: AWS (EC2, S3, Lambda, IAM, VPC, CloudWatch), OpenAI API, LLM Integration, Prompt Engineering

DevOps & Tools: Git, GitHub Actions, Docker, Vercel, Render, Postman, Figma, CI/CD Pipelines

Concepts: System Design, DSA, OOP, Agile/Scrum, Cloud Security, Scalable Architecture

PROFESSIONAL EXPERIENCE

Software Developer Intern — Sparks to Ideas

2023

- Engineered and maintained production-grade React components consumed across the live platform, improving UI consistency, render performance, and overall user experience.
- Architected optimized state management patterns using React Hooks and Redux, eliminating redundant re-renders and improving data throughput for large payload responses.
- Developed a reusable, well-documented component library adopted across the codebase, accelerating feature delivery and enforcing design system consistency.
- Collaborated within an Agile sprint workflow, contributing to code reviews, daily stand-ups, and sprint retrospectives alongside senior engineers.

PROJECTS

DPI — Deep Packet Inspector

Java · Node.js · Express.js · React.js · TailwindCSS · Multithreading

- Built an end-to-end Deep Packet Inspection engine that parses PCAP files, classifies network traffic by application layer, and applies an AI layer to convert raw output into executive-ready analyst summaries for cybersecurity auditing and network observability pipelines.
- Reduced processing latency from 60s to under 3s (~95% improvement) by re-engineering the multi-threaded worker lifecycle with max-packet controls and optimized thread shutdown logic.
- Delivered a React.js dashboard with real-time flow statistics, an AI-powered insights panel, and one-click downloadable reports.

ContentForge — AI Marketing Content Platform

Next.js · OpenRouter AI · TailwindCSS · REST API · Prompt Engineering · Vercel

- Engineered a full-stack AI platform that converts a single campaign brief into ready-to-publish copy across 7 platforms (Instagram, LinkedIn, Email, Ads), cutting content drafting time by ~60%.
- Implemented tone-based rewriting, A/B variant generation, and audience targeting — fully deployed on Vercel and designed for non-technical marketing users.

CropCart — MERN Stack Marketplace Platform

React.js · Node.js · Express.js · MongoDB · JWT · Redux · bcrypt

- Built a full-stack two-sided marketplace connecting farmers and consumers, featuring role-based dashboards, messaging, order management, and an admin panel.
- Secured with JWT authentication, bcrypt password hashing, and persistent MongoDB storage for reliable, production-ready data handling.

EDUCATION

B.Tech in Computer Science Engineering — Karnavati University

2024 – Present

Ahmedabad, Gujarat

Diploma in Information Technology — LJ University

2021 – 2024

CGPA: 7.50 / 10

Secondary Education (SSC) — Nelson's High School

2019

Score: 85.11%

CERTIFICATIONS

AWS Academy Graduate — Cloud Security Foundations

2026

AWS Academy Graduate — Cloud Foundations

2025